

Technical Documentation & Process Guide — Version 2.0

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What's new in Version 2.0

Major architectural restructure: FO Number replaces Delivery Order Number as the primary key. One FO can now cover multiple DO numbers and multiple distributors. Two SharePoint lists replace one. Two-stage payment system (initial + final). Four automated flows. Teams notifications at every milestone.

4
Flows

4
Forms

2
SP Lists

5
Stakeholders

2
Payments

Table of Contents

1. Project Overview & Background	3
2. What Changed in Version 2.0	3
3. System Architecture	4
4. SharePoint Lists	5
5. Microsoft Forms	7
6. Flow 1 — Invoice Submission	9
7. Flow 2 — FO Number & FC Approval	10
8. Flow 3 — Initial Payment Confirmation	11
9. Flow 4 — Delivery Confirmation & Final Payment	12
10. Teams Notifications	13
11. End-to-End Process Lifecycle	14
12. Error Handling & Troubleshooting	15
13. Enhancement Roadmap	15

1. Project Overview & Background

Background

MPL Ghana's logistics operations require a structured process for tracking transporter invoices from submission through to final payment. This system automates the entire lifecycle — from the Transport Manager's initial submission through Finance Controller approval to two-stage payment processing — eliminating manual handoffs, creating a full audit trail, and notifying all stakeholders via both email and Microsoft Teams.

Stakeholders

Role	Responsibility	Tools Used
Transport Manager	Submits invoice with multiple DO numbers, confirms delivery, triggers final payment	Form 1, Form 4
Tracking Manager	Generates unique FO Number linking all DOs	Form 2
Finance Controller (FC)	Reviews and approves or rejects payment via email buttons	Approval email
Payment Office	Processes initial and final payments, confirms via forms	Form 3
System (Power Automate)	Orchestrates all handoffs, updates SharePoint, sends notifications	4 Flows

2. What Changed in Version 2.0

Area	Version 1.0	Version 2.0
Primary Key	Delivery Order Number (DO)	FO (Freight Order) Number — generated by Tracking Manager
Multiple DOs	Not supported — one DO per submission	Fully supported — comma-separated DOs parsed into individual rows
SharePoint Lists	One list (Invoice Tracker)	Two lists: FO Master + DO Details (relational structure)
Distributors	Maximum 2 distributors	Unlimited distributors — one row per DO in DO Details
Payments	Single payment	Two-stage: Initial Payment + Final Payment after delivery confirmation
Delivery Confirmation	Not required	Transport Manager confirms delivery before final payment triggers
Forms	3 forms	4 forms — new Form 4 for delivery confirmation and final payment
Flows	2 flows	4 flows — new Flow 3 and Flow 4 for two-stage payment
Notifications	Email only	Email + Microsoft Teams (#Payment Alerts channel)
Status Tracking	5 statuses	9 statuses covering full two-payment lifecycle

3. System Architecture

End-to-End Process Flow

Stage	Actor	Action	SP Status After
1	Transport Manager	Submits Form 1 with multiple DOs, distributor names, Invoice + Waybill PDFs	Pending Tracking
2	Flow 1	Creates FO Master row. Parses DO numbers. Creates one DO Details row per DO. Emails Tracking Manager	Pending Tracking
3	Tracking Manager	Generates unique FO Number. Submits Form 2	Pending Tracking
4	Flow 2	Updates FO Master and all DO Details rows with FO Number. Sends FC approval email with all details	Pending FC Approval
5a	Finance Controller	Clicks APPROVE in email	Initial Payment Pending
5b	Finance Controller	Clicks REJECT in email	Rejected
6	Flow 2 YES branch	Updates SP. Emails Payment Office + Transport Manager. Teams notification	Initial Payment Pending
7	Payment Office	Processes initial payment. Submits Form 3 with reference and payment advice	Initial Payment Pending
8	Flow 3	Updates FO Master with payment details. Emails FC + Transport Manager with Form 4 link. Teams notification	Initial Payment Made
9	Transport Manager	Confirms delivery via Form 4 link in email. Enters final payment amount	Initial Payment Made
10	Flow 4	Checks delivery confirmed. Updates SP. Emails Payment Office + FC + Transport Manager. Teams notification	Final Payment Pending
11	Payment Office	Processes final payment	Final Payment Made → Complete

4. SharePoint Lists

Site: <https://tolaramsg.sharepoint.com/sites/MPLGhana>

Why two lists?

One FO Number covers multiple Delivery Orders and multiple distributors. A single flat list cannot represent this one-to-many relationship cleanly. FO Master holds one row per FO — all vendor, payment, and approval data. DO Details holds one row per DO — linked back to FO Master via FO Number.

List 1 — FO Master

Primary key: FO Number (indexed, unique values enforced)

Column	Type	Required	Notes
FO Number	Single line	Yes	PRIMARY KEY — indexed, unique values enforced
Transporter Name	Single line	Yes	
Transporter Vehicle Number	Single line	Yes	
Vendor Account Name	Single line	Yes	Bank/MoMo
Vendor Account Number	Single line	Yes	Bank/MoMo
Invoice Number	Single line	Yes	Used as matching key before FO Number is generated
Invoice PDF URL	Hyperlink	Yes	Extracted from Forms JSON upload response
Waybill PDF URL	Hyperlink	Yes	Extracted from Forms JSON upload response
Amount to be Paid to Transporter	Single line	Yes	
FC Decision	Choice	No	Pending / Approved / Rejected
FC Comments	Multi-line	No	FC approval/rejection reason
Status	Choice	No	See status lifecycle below
Payment Office Notified	Yes/No	No	Default: No
Delivery Confirmed	Yes/No	No	Default: No — set by Flow 4
Initial Payment Ref	Single line	No	Set by Flow 3
Initial Payment Amount	Single line	No	Set by Flow 3
Initial Payment Date	Date/Time	No	Set by Flow 3
Initial Payment Advice URL	Hyperlink	No	Set by Flow 3
Final Payment Amount	Single line	No	Set by Flow 4

Final Payment Ref	Single line	No	Set by final payment confirmation
Final Payment Date	Date/Time	No	Set by final payment confirmation
Final Payment Advice URL	Hyperlink	No	Set by final payment confirmation
Payment Completed Date	Date/Time	No	Set when Status = Complete

List 2 — DO Details

Foreign key: FO Number (links to FO Master). One row per Delivery Order.

Column	Type	Required	Notes
Title	Single line	No	Stores Invoice Number temporarily for matching in Flow 2
FO Number	Single line	No	Foreign key — written by Flow 2 after Tracking Manager submits
DO Number	Single line	Yes	Individual Delivery Order number
Distributor Name	Single line	Yes	Distributor receiving this DO
Distributor Code	Single line	No	Distributor code

Status Column Lifecycle

Status	Set By	Meaning
Pending Tracking	Flow 1	Form 1 submitted. Waiting for Tracking Manager to generate FO Number
Pending FC Approval	Flow 2	FO Number generated. Waiting for Finance Controller decision
Initial Payment Pending	Flow 2 YES branch	FC approved. Payment Office notified for initial payment
Initial Payment Made	Flow 3	Initial payment processed and confirmed
Final Payment Pending	Flow 4	Delivery confirmed. Payment Office notified for final payment
Final Payment Made	Manual/Flow	Final payment processed
Complete	Manual/Flow	Both payments completed — transaction closed
Rejected	Flow 2 NO branch	FC rejected the invoice

5. Microsoft Forms

Form 1 — Transport Manager Invoice Submission

Form name: *Transport Manager – Invoice Submission*

#	Question	Type	Required	Notes
1	Transporter Name	Text	Required	
2	Transporter Vehicle Number	Text	Required	
3	Vendor Account Name (Bank/MoMo)	Text	Required	
4	Vendor Account Number (Bank/MoMo)	Text	Required	
5	Invoice Number	Text	Required	Used as matching key before FO Number exists
6	Delivery Order Numbers	Text	Required	Comma-separated. e.g. DO-001, DO-002, DO-003
7	Distributor Names	Text	Required	Comma-separated, same order as DOs. e.g. Cepodek, Asana
8	Amount to be Paid to Transporter	Text	Required	
9	Attach Invoice (PDF)	File Upload	Required	PDF only, max 1 file
10	Attach Waybill (PDF)	File Upload	Required	PDF only, max 1 file

Form 2 — Tracking Manager FO Number Entry

Form name: *Tracking Manager – FO Number Entry*

#	Question	Type	Required	Notes
1	Transport Manager Name	Text	Required	Used to link back to Form 1 submission
2	Invoice Number	Text	Required	Must match Form 1 Invoice Number exactly
3	FO (Freight Order) Number	Text	Required	PRIMARY KEY — Tracking Manager generates this unique number

4	Total Number of DOs	Number	Required	Total count of Delivery Orders under this FO
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Form 3 — Initial Payment Confirmation

Form name: *Payment Office – Payment Confirmation*

#	Question	Type	Required	Notes
1	FO Number	Text	Required	Must match FO Number from Form 2 exactly
2	Payment Reference Number	Text	Required	
3	Amount Paid	Text	Required	
4	Date of Payment	Date	Required	
5	Payment Advice / Screenshot	File Upload	Required	PDF or image, max 1 file

Form 4 — Delivery Confirmation & Final Payment Request

Form name: *Transport Manager – Delivery Confirmation & Final Payment*

#	Question	Type	Required	Notes
1	FO Number	Text	Required	From initial payment confirmation email
2	Delivery Confirmation	Choice	Required	Yes — Delivery completed / No — Not completed
3	Delivery Confirmation Notes	Text	Optional	Any notes about delivery condition
4	Final Payment Amount	Text	Required	Amount to be paid after delivery
5	Any Deductions or Adjustments?	Choice	Required	Yes — There are deductions / No — Pay full amount
6	Deduction Amount	Text	Optional	Amount to deduct from final payment
7	Deduction Reason	Text	Optional	Reason for deduction

6. Flow 1 — Invoice Submission

Trigger: When a new response is submitted (Form 1)

Step	Action	Connector	Details
Trigger	When a new response is submitted	Microsoft Forms	Fires when Transport Manager submits Form 1
1	Get response details	Microsoft Forms	Retrieves all form answers using Response Id
2	Initialize variable — DONumbersArray	Variable	split(DO Numbers field, ',') → Array of individual DO numbers
3	Initialize variable — DistributorNamesArray	Variable	split(Distributor Names field, ',') → Array of distributor names
4	Initialize variable — LoopCounter	Variable	Integer starting at 0. Used to match DO to Distributor by index
5	Create item — FO Master	SharePoint	Creates new row in FO Master with all Form 1 data. Status = Pending Tracking. Invoice PDF and Waybill PDF extracted using first(json(...))?[link] expression
6	Apply to each — DONumbersArray	Control	Loops through each DO number in the array
6a	Create item — DO Details	SharePoint	Creates one row per DO. DO Number = trim(item()). Distributor Name = trim(split(Distributor Names, ',')?[LoopCounter])
6b	Increment variable — LoopCounter	Variable	Adds 1 to counter so next iteration picks the matching distributor name
7	Send email — Tracking Manager	Office 365 Outlook	Email with all invoice details and clickable link to Form 2

Key Technical Notes

- File uploads return JSON arrays — use first(json(body('Get_response_details')?[FIELD_KEY]))?[link] to extract URL
- Invoice PDF field key: re78d3fcfb3fc4f9bbe4eec3f1f456cbd
- Waybill PDF field key: rd958bc8c4f6943c191fb96fd7d817398
- DO Numbers field key: r5002011ddb09454897f72b165ca47b47
- Distributor Names field key: r7eaeecf573e547eca46c9e3392fecb23
- The LoopCounter variable ensures each DO is matched to its corresponding distributor name by array index
- Title column in FO Master is set to Invoice Number for easy identification in the list view

7. Flow 2 — FO Number & FC Approval

Trigger: When a new response is submitted (Form 2)

Step	Action	Connector	Details
Trigger	When a new response is submitted	Microsoft Forms	Fires when Tracking Manager submits Form 2
1	Get response details	Microsoft Forms	Retrieves FO Number, Invoice Number, Transport Manager Name
2	Get items — FO Master	SharePoint	Filter: InvoiceNumber eq '[Invoice Number]'. Top Count: 1
3	Get items — DO Details	SharePoint	Filter: Title eq '[Invoice Number]'. Top Count: 10. Gets all DO rows for this invoice
4	Apply to each — DO Details	Control	Updates every DO Details row with the new FO Number
4a	Update item — DO Details	SharePoint	Writes FO Number to each DO row
5	Apply to each — FO Master	Control	Updates FO Master and runs approval
5a	Update item — FO Master	SharePoint	Writes FO Number. Status = Pending FC Approval
5b	Start and wait for approval	Approvals	Sends structured approval email to FC (Evans, Ajit, Jerry) with all invoice details, DO summary, PDF links
5c	Condition — Outcome = Approve	Control	Branches based on FC decision
TRUE 1	Update item — FO Master	SharePoint	FC Decision = Approved. Status = Initial Payment Pending
TRUE 2	Send email — Payment Office	Office 365 Outlook	Full invoice details with PDF links. Request initial payment
TRUE 3	Send email — Transport Manager	Office 365 Outlook	Invoice approved notification
TRUE 4	Post message — Teams	Microsoft Teams	Notifies #Payment Alerts channel of FC approval
FALSE 1	Update item — FO Master	SharePoint	FC Decision = Rejected. FC Comments written. Status = Rejected
FALSE 2	Send email — Transport Manager	Office 365 Outlook	Rejection notification with FC comments

FALSE 3	Post message — Teams	Microsoft Teams	Notifies #Payment Alerts channel of rejection
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8. Flow 3 — Initial Payment Confirmation

Trigger: When a new response is submitted (Form 3)

Step	Action	Connector	Details
Trigger	When a new response is submitted	Microsoft Forms	Fires when Payment Office submits Form 3
1	Get response details	Microsoft Forms	Retrieves FO Number, Payment Ref, Amount, Date, Payment Advice PDF
2	Get items — FO Master	SharePoint	Filter: FONumber eq '[FO Number]'. Top Count: 1
3	Apply to each — FO Master	Control	Updates FO Master with initial payment details
3a	Update item — FO Master	SharePoint	Writes Initial Payment Ref, Amount, Date, Advice URL. Status = Initial Payment Made. Payment Office Notified = Yes
3b	Send email — Finance Controller	Office 365 Outlook	Payment confirmation with all payment details and advice PDF link
3c	Send email — Transport Manager	Office 365 Outlook	Payment confirmation with Form 4 link embedded as clickable button for delivery confirmation
3d	Post message — Teams	Microsoft Teams	Notifies #Payment Alerts: Initial payment confirmed with FO Number and amount

Key Technical Notes

- FO Number field key: r7a7078918c6842b3a82c89a8fa37d26e
- Payment Reference field key: r1a4d2aa86dea4fd9b0ec0196f8e70b4b
- Amount Paid field key: rd589d6b651fe4567aa1820cac242adc4
- Date of Payment field key: r869b7915b1134d26904eb71b2d81006b
- Payment Advice PDF field key: rdb210f45625f4da587ed9489cec7ae72
- The Form 4 link in the Transport Manager email is the key trigger for the two-stage payment — Transport Manager clicks it only after delivery is completed
- Payment Advice URL extracted using:
first(json(body('Get_response_details')?['rdb210f45625f4da587ed9489cec7ae72']))?['link']

9. Flow 4 — Delivery Confirmation & Final Payment

Trigger: When a new response is submitted (Form 4)

Step	Action	Connector	Details
Trigger	When a new response is submitted	Microsoft Forms	Fires when Transport Manager submits Form 4
1	Get response details	Microsoft Forms	Retrieves FO Number, Delivery Confirmation, Notes, Final Amount, Deductions
2	Condition — Delivery Confirmed?	Control	Checks if Delivery Confirmation field contains 'Yes'
FALSE	Send email — Transport Manager	Office 365 Outlook	Notifies that delivery not confirmed — no payment triggered. Asks to resubmit when delivered
TRUE 1	Get items — FO Master	SharePoint	Filter: FONumber eq '[FO Number]'. Top Count: 1
TRUE 2	Apply to each — FO Master	Control	Updates FO Master and sends final payment notifications
TRUE 2a	Update item — FO Master	SharePoint	Delivery Confirmed = Yes. Final Payment Amount written. Status = Final Payment Pending
TRUE 2b	Send email — Payment Office	Office 365 Outlook	Full details: FO Number, transporter info, delivery confirmation, final amount, deductions, initial payment reference, PDF links
TRUE 2c	Send email — Transport Manager	Office 365 Outlook	Confirms final payment request sent to Payment Office
TRUE 2d	Send email — Finance Controller	Office 365 Outlook	FYI notification that final payment has been triggered
TRUE 2e	Post message — Teams	Microsoft Teams	Notifies #Payment Alerts: Final payment triggered with FO Number and amount

Key Technical Notes

- FO Number field key: r9fdc19b3929c49cdaeb9825cc3f3d486
- Delivery Confirmation field key: r8168a59ea79443d0961e6edd2cd0b4cf
- Final Payment Amount field key: r3153f9a842484c58869bbbe52261e888
- Deduction field key: r2df3cacf5c4b42209d01894c0f18e893
- Condition checks if delivery field CONTAINS 'Yes' — not equals — to handle the full choice text
- If delivery is not confirmed, flow stops at FALSE branch — no payment is triggered
- Transport Manager can only submit Form 4 via the link in the initial payment email — this ensures payment sequence is respected

10. Teams Notifications

Channel: #Payment Alerts | Connector: Post a message in a chat or channel (Microsoft Teams)

Trigger Point	Flow	Message Content
FC Approves invoice	Flow 2 YES branch	FO Number, Invoice No, Transporter, Amount, Status: Initial Payment Pending
FC Rejects invoice	Flow 2 NO branch	FO Number, Invoice No, Transporter, Amount, Rejection reason
Initial payment confirmed	Flow 3	FO Number, Invoice No, Payment Ref, Amount Paid, Date
Final payment triggered	Flow 4	FO Number, Invoice No, Transporter, Final Amount, Deductions, Delivery confirmed

11. End-to-End Process Lifecycle

#	Who	What	How	Result
1	Transport Manager	Submits invoice with multiple DOs	Form 1	FO Master + DO Details rows created
2	System (Flow 1)	Notifies Tracking Manager	Email with Form 2 link	Tracking Manager alerted
3	Tracking Manager	Generates FO Number	Form 2	FO Number submitted
4	System (Flow 2)	Updates SP + sends FC approval	SharePoint + Approvals	FC receives Approve/Reject email
5	Finance Controller	Approves or rejects	Approval email buttons	Decision recorded in SP
6	System (Flow 2)	Notifies Payment Office or TM	Email + Teams	Payment Office alerted OR TM notified of rejection
7	Payment Office	Makes initial payment	Form 3	Initial payment confirmed
8	System (Flow 3)	Updates SP + sends Form 4 link to TM	Email + Teams	TM receives delivery confirmation link
9	Transport Manager	Confirms delivery + requests final payment	Form 4 link in email	Final payment triggered
10	System (Flow 4)	Notifies Payment Office for final payment	Email + Teams	Payment Office alerted for final payment
11	Payment Office	Makes final payment	Manual/Form	Transaction complete

12. Error Handling & Troubleshooting

Error	Cause	Fix
Column 'FONumber' does not exist	Internal column name mismatch	Check column URL in SP List Settings — use exact internal name from Field= parameter
FO Number is required	Flow 1 runs before FO Number exists	Set FO Number as NOT required in SharePoint — Flow 2 writes it later
Get items returns 0 results	Invoice Number in Form 2 doesn't match Form 1 exactly	Copy Invoice Number from SP list — enter exactly in Form 2
Distributor name not matching DO	LoopCounter not incrementing	Check Increment variable action is inside Apply to each after Create item
File upload returns JSON error	Raw upload token used instead of expression	Use: <code>first(json(body('Get_response_details')?['KEY']))?['link']</code>
Approval email not received	Approvals connector sign-in expired	Go to Flow → Edit → Approval action → Change connection → Sign in again
Condition always goes to FALSE	Condition placed outside For each loop	Move Condition inside the same Apply to each as the approval action
Flow 4 not triggering final payment	Delivery Confirmation answer doesn't contain 'Yes'	Condition must use 'contains' not 'equals' — full choice text is longer than just Yes
Teams message not posting	Teams connector sign-in or channel permissions	Re-authenticate Teams connector — check bot has permission to post in channel

13. Enhancement Roadmap

Enhancement	Status	Details
Multiple DOs per FO	Completed v2.0	Comma-separated DOs parsed into individual DO Details rows using split() and LoopCounter
FO Number as primary key	Completed v2.0	Replaces Delivery Order Number — generated by Tracking Manager
Two-stage payment	Completed v2.0	Initial payment via Flow 3, final payment via Flow 4 after delivery confirmation
Delivery confirmation gate	Completed v2.0	Transport Manager must confirm delivery before final payment triggers
Teams notifications	Completed v2.0	All payment milestones posted to #Payment Alerts channel
FC approval notification to TM	Completed v2.0	Transport Manager notified when FC approves
24-hour FC reminder	Planned	Set approval timeout to PT24H — send reminder if FC hasn't responded
Power BI dashboard	Planned	Connect Power BI to FO Master and DO Details lists. Visuals: status breakdown, amounts by distributor, payment timeline, DO count per FO
Final payment confirmation flow	Planned	Flow 5 triggered when Payment Office confirms final payment — updates Status to Complete
Power Apps form	Future	Replace Microsoft Forms with Power Apps for richer UI — pre-fill fields, show existing SP data to users